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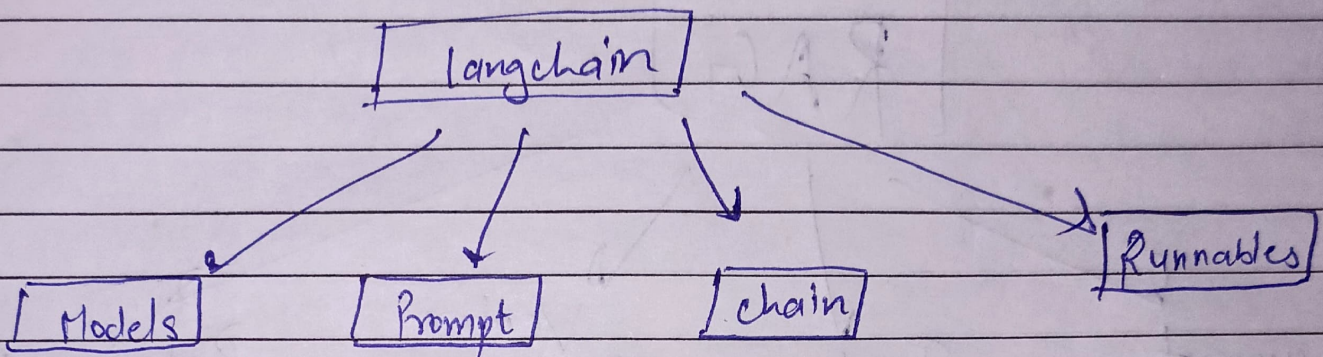
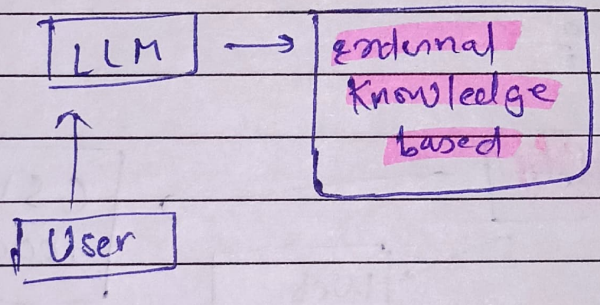
Document loaders in langchain :-

Plan of Action



- RAG → (Retrieval Augumented Generation)



- RAG is a Technique that combines Information retrieval with language generations, where a model retrieves relevant documents from a knowledge base and then uses them as content to generate accurate and grounded responses.

Benefits of using RAG :-

1. Use of up to date Information
2. Better privacy
3. No limit of documents size.

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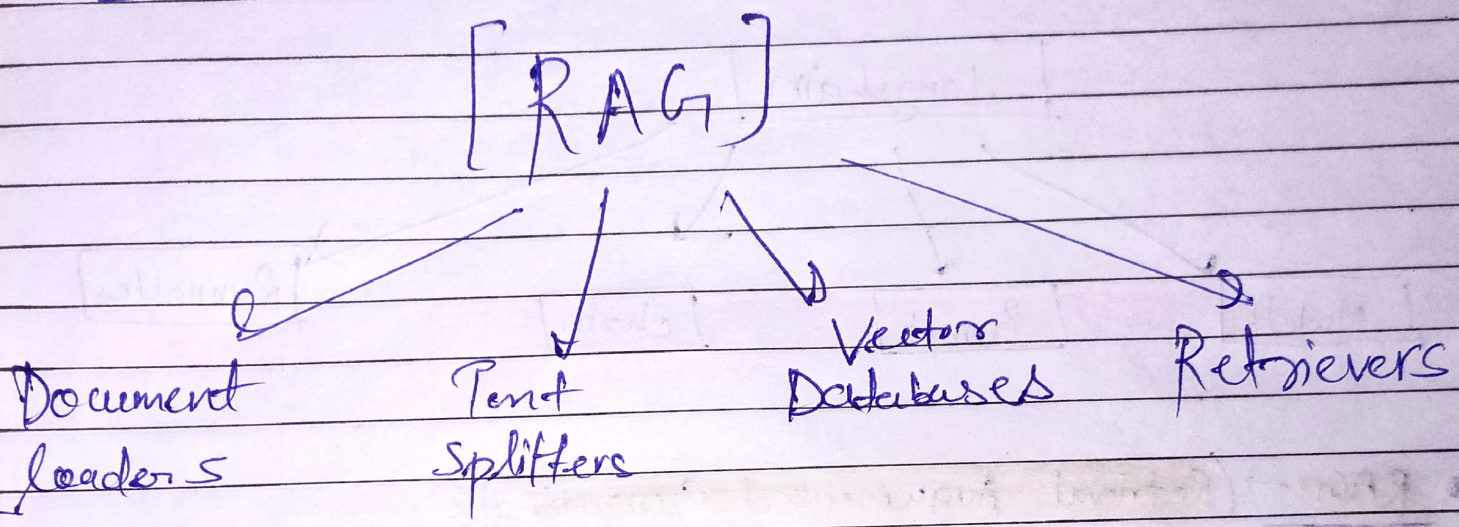
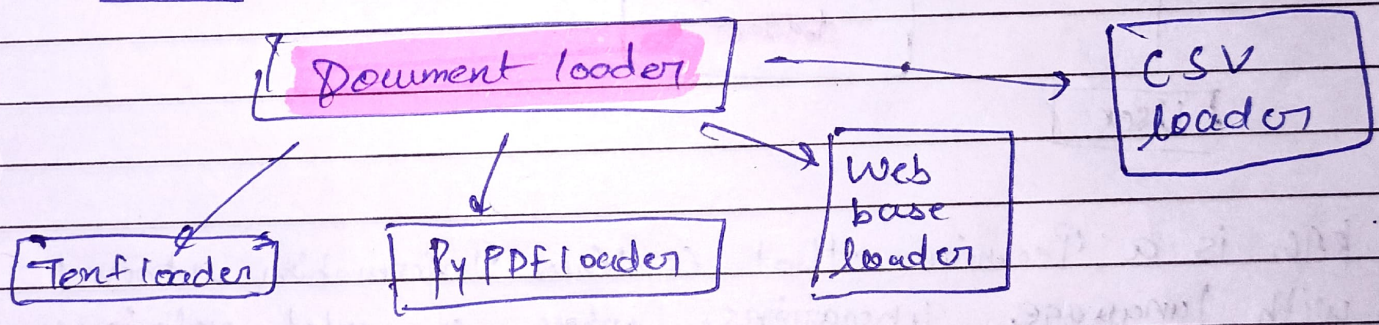
#RAG Components :-



(1) Document loaders :-



Definition → Document loader are components in langchain used to load data from various sources into a standardized format (usually as document object) which can then be used for chunking embeddings, retrieval and generation.

Document (

Page_Content = "the acutual text Content",
metadata = {"source": "filename.pdf", ...}

)

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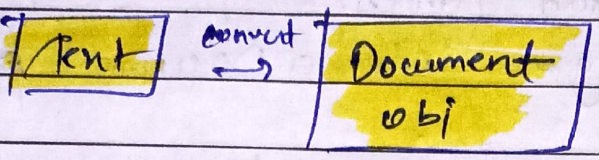
- **TextLoader** :- is a simple and commonly used document loader in langchain that reads plain text (.txt) files and convert them into a langchain document object.



- Ideal for loading chat logs, code shippets, scraped text, plain text data into a langchain pipelines.
- work only on Text files (.Txt)

doc loader = [ ]
doc list me!!

```
from langchain_community.document_loaders import TextLoader
loader = TextLoader("file path", encoding='utf-8')
docs = loader.load()
print = (type(docs)) → Type - list
print = (len(docs)) → 1 → Kitne document load kiye hai
print = (doc[0]) → Poora Document object print karega
print = (docs[0].page_content) → actual text
print = (docs[0].metadata) → Extra info.
```

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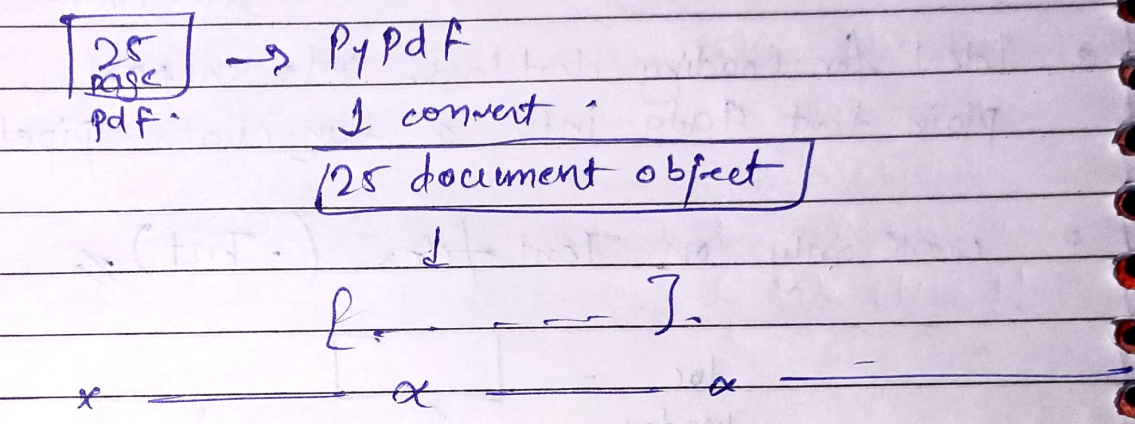
② **PyPDF loader** :- is a document loader in langchain used to load content from pdf files and convert each pages into a Document object



Document (page_content = 'text from page 1', metadata = {"page": 0, "source": "file.pdf"}),

. . .

)

limitations :- it uses the pyPDF library under the hood
- not great with scanned PDFs or complex layouts.

pyPDF loader → not good working with Scanned pdf

Use Cases	Recommended loaders
Simple, clean pdf	PyPDF loader
PDFs with Tables / columns	PDF Plumber Loader
Scanned / Image PDFs	Unstructured PDF loader or Amazon Textract PDF loader
Need layout And Image data	PyMUPDF loader
Want best Structure extraction	Unstructured PDF loader.

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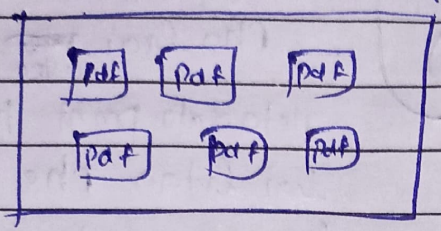
③ Directory Loader → load multiple documents from a directory of files.

Folder Mai Pdf



Glob pattern	What it loads
"**/*.txt"	All .txt files in all subfolders
"*.pdf"	All .pdf files in the root directory
"data/*.csv"	All .csv files in the data/ folder
"**/*"	All files (any type, all folder)

[** = recursive search through subfolders]

④ Load v/s lazy load

load
↓
eager loading

- A list of document obj
- load all ~~memory~~ documents immediately into memory
- Best when
  - → the number of documents is small.
  - → you want everything loaded up front.

lazy load
↳ generator of docs

- Document are not all loaded at once, they fetched one at a time as needed.
- Best when:
  - you are dealing with large documents or lots of files.
  - you want to stream processing. (eg. chuncking, embedding) without

Spiral

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lazy-load :

Date .....

```
docs = loader.lazy_load()

for document in docs:
    print(document.metadata)
```

Note :

Har bar memory mai ek document aa rha hai ~~wala~~ uska hum Metadata print krr rhe hai hata rhe hai fir agla wala aa rha hai uska Metadata print krr rhe hai aur aisa hi hota rhega !!

---

(5) **Web Base loader** : is a document loader in langchain used to load and extract text content from web pages (URLs).

~~Request~~

or

- It ~~uses~~ Beautiful soup under the hood to parse HTML and extract visible text.
- When to use → for blogs, news article, or public websites, where text based content.
- limitations :-
  - Doesn't handle JavaScript heavy pages well (use Selenium URL Loader for that)
  - Load only static content (what is in the HTML not what loads after the page renders).

Date .....

(6) **CSV_loader** → is a document loader used to load
- CSV files into langchain Documents objects - one per row, by default.

same code on github - - - -